Uneven Bars?
An Analysis of Social Pressure in NCAA Women's Gymnastics

Seth Cannon, Tommy Morgan, Eli Morse

Brigham Young University

Abstract

Theories on social pressure suggest that the people in an actor's immediate environment can impact their performance and decision-making. This study utilizes a unique dataset of scores from NCAA women's gymnastics meets to examine whether Black gymnasts experience intergroup anxiety, a type of negative social pressure induced by the anticipation of interacting with members of an outgroup. We use a fixed-effects regression model to analyze the full regular seasons of every team that visited Brigham Young University (BYU) between the 2017 and 2022 seasons and find no evidence that being Black while performing at BYU induces a negative effect on scores in any of the four women's gymnastics events or in scoring overall. However, we fail to definitively conclude that these gymnasts do not experience any amount of intergroup anxiety due to an imbalanced panel.

Section I. Introduction

The racial history of women's gymnastics is unique in that past absence of Black gymnasts is not uniquely attributable to explicit racial discrimination. The extensive training required to excel in gymnastics incurs investments of time and resources that usually restrict participation to children in relatively wealthy households, and relatively wealthy households in the United States are relatively unlikely to be Black households. Though gymnasts like Simone Biles and Jordan Chiles are world-class competitors today, Black gymnasts are still well in the minority of gymnasts at the college level (NCAA 2018). This being the case, does being in the minority at meets harm those Black gymnasts' performance?

In the world of sports, much literature exists on the effects of *explicit* interracial interactions, such as referees of one race calling personal fouls on basketball players of another race (Price & Wolfers 2010). There is also literature describing how audiences can exert pressure that affects game-altering decisions, like when referees add more stoppage time to the ends of soccer games in which the home team is down by one goal (Garicano, Palacios-Huerta, & Prendergast 2005). Our study aims to identify an *implicit* interracial interaction between a gymnast and the audience at a meet known as "intergroup anxiety," the ambiguous feeling of discomfort or stress that arises when a person is faced with having to interact with people from another group (Stephan & Stephan, 1985). For reasons we describe in Section II, this study aims to uncover any effect of intergroup anxiety on a gymnast by examining whether there are heterogeneous effects of performing at BYU for Black and non-Black gymnasts.

We analyze the performance of NCAA gymnasts of all races who competed at BYU over the full span of meets in the six regular seasons from 2017-2022 and find no evidence for the presence of intergroup anxiety in Black gymnasts. Section II gives background information on

NCAA women's gymnastics meets, intergroup anxiety, and our dataset, Section III discusses our experimental design and identification strategy, Section IV reports the results of our analysis, and Section V concludes.

Section II. Background

In gymnastics, a meet is composed of four events: vault, floor, balance beam, and uneven bars. Gymnasts typically compete in only one or two of these events, but each team has at least one "all-around" gymnast who does all four. Each event is scored out of 10 by two judges whose independent scores of each performance are averaged to a final performance score. The typical regular season meet has four judges – two from in region and two from out of region – each judging two events, with two judges on each event. When there are more than two teams at a given meet, eight individual judges judge one event each with no rotation between events.

In each event, six gymnasts from each team perform, and the lowest of those six scores is dropped. After all events are complete, the sum of the five highest scores in each event are summed to compute the team's meet score. Though an individual score can theoretically range anywhere from zero to 10, they are clustered very tightly within the 9.5 - 9.9 range. Individual performance scores are summed across the four events to compute the meet score. Because the practical range of scores is so small, tiny differences in average scores separate elite teams from great and decent teams. Specifically, elite gymnastics teams have the potential to hit a 198.00 meet score, which is obtainable only with an average score of 9.9 from every gymnast in every event across the entire meet. Great teams can consistently hit a 197.00 meet score (a 9.85 average performance score), and good teams can consistently hit a 196.00 meet score (a 9.8 average performance score).

Each of the four gymnastics events is a high-intensity, high-focus exercise. We theorize that any additional outside pressures could cause gymnasts to make unintentional mistakes they would not make absent such pressure. Potential pressures of this kind include Intergroup anxiety, the ambiguous feelings of discomfort or stress that arise when a person is faced with having to interact with people from another group. This effect was first identified by Stephan and Stephan (1985), and further investigation suggests that a history of negative relations between groups predicts greater intergroup anxiety (Britt et. al. 1996). Given that there is a history of inter-university boycotts and widely published opinion pieces over racial inequality within BYU's athletic programs, we argue that if a Black gymnast would experience intergroup anxiety anywhere, she would experience it at BYU.

The combination of apparently different distributions in scoring among gymnasts of different race groups at BYU and the possible presence of social pressures like intergroup anxiety motivate this analysis. Our study uses hand-compiled data from the official meet reports submitted by judges at the end of each meet. These report sheets contain information on each gymnasts' score(s) at a meet, the judges for each event, and the venue of the meet. The majority of those sheets are found at RoadToNationals.com, the official statistical website of NCAA Gymnastics. Any sheets not found on RoadToNationals were pulled from individual team websites. We coded the race of each gymnast based on visual inspection of their official photographs as posted on each school's official gymnastics website. The data we collected is well-suited for our analysis because it allows us to analyze scores across race groups and meet venues. Because variation in scores is still small and the "perfect 10" upper bound can give undue weight to exceptionally low scores, we drop scores below the third percentile for each event. Kernel density estimations of score distributions by event (Figure 1) show that scoring in

vault and floor is relatively consistent between race groups and venues. Scoring distributions in beam and bars appear to be less consistent across the same groups. Full-sample distributions of scores across events appear to differ as well.

Our analysis is hindered by weaknesses of our dataset, including that our data was hand-entered and at risk of input error; however, rigorous verification strategies employed in the collection phase mitigated this risk. More concerning is that our dataset is unbalanced: scores received by Black gymnasts at BYU make up less than two percent of all scores in our dataset (see Table 1 for full summary stats). Running our model with such a small sample in the "treated group" likely gives undue weight to outlier scores and obfuscates our results; we discuss these implications further in Section IV.

Section III. Experimental Design

Our analysis uses a fixed effects regression with the goal of isolating the heterogeneous effects of competing at BYU for Black and non-Black gymnasts. The data used in our analysis allows us to control for gymnast, venue, year, and judge-specific characteristics that might influence the score a gymnast receives at a given meet. In Table 2 we report the results from estimating

$$(1)\quad \text{Score}_{ivjy} = \beta(\text{Black x atBYU})_{iv} + [\text{gymnast effects}]_i + [\text{venue effects}]_v + [\text{judge effects}]_j + [\text{year effects}]_y + [\text{gymnast x year effects}]_{iy} + [\text{judge x race effects}]_{ij} + \varepsilon_{ivjy}$$

where subscripts *i, v, j,* and *y* refer to individual gymnasts, venues, judges, and years, respectively. The dependent variable is the score earned by a gymnast in a given event. Each of

the square brackets denote a set of fixed effects, and the interaction term (Black x atBYU)$_{iv}$ takes a value of 1 if a score is received by a Black gymnast competing at BYU and 0 otherwise. The coefficient of interest is β, which we interpret as the differential impact of competing at BYU on Black gymnasts relative to non-Black gymnasts. To account for possible correlation between the scores received by a given gymnast, we report standard errors clustered at the gymnast level.

In the first row of Table 2, we report the coefficients on the interaction term from regressions in each event of score on binary variables for being Black and being at BYU and our interaction term. In the second row, we replace the binary variables with gymnast and venue fixed effects, which control for observable and unobservable characteristics of a given gymnast or venue that do not change between meets, like a gymnast's upbringing or a venue's altitude. In the third row, we add judge and year fixed effects to control for things like where a given judge is from or a worldwide pandemic that begins in a given year. In the final row, we add gymnast by year and judge by race fixed effects, which control for a given gymnast's seniority on the team, a given judge's implicit racial bias, and other similar factors. When the regression includes scores from all events, we also add event fixed effects to the third and fourth row models.

We assume that including the fixed effects described above isolates the differential effect of competing at BYU on Black versus non-Black gymnasts. For example, given the potential presence of racist judges, we add judge-by-race fixed effects to capture the attitudes a judge holds about gymnasts that do not change across races. We find it difficult to think of any significant source of scoring variance not captured by our set of fixed effects or the interaction term of interest. As such, we assume that – after controlling for these fixed characteristics – our model fulfills the zero conditional mean assumption.

Section IV. Results

From our analysis, we find that the coefficient on the interaction term (Black x atBYU) is not statistically different from zero in any individual event or overall after applying our full set of fixed effects. Though there sometimes appears to be a clear visual difference in the score densities between race categories, like on the uneven bars (see Figure 1), our analysis reveals that such a difference is not statistically significant. This means that competing at BYU does not seem to have an adverse effect on Black gymnasts' performance, which suggests that they are not affected enough by intergroup anxiety or any other implicit social pressures to suffer a drop in performance at BYU.

Between possible assumptions about BYU because of its history and the likely demographics of its audience members, we believed that the potential for Black gymnasts to be more racially self-aware was higher and that competing at BYU would cause them to experience race-related intergroup anxiety. Whether any or all of the Black gymnasts we analyze experience such pressure or not, we find no evidence that such a pressure negatively affects the performance or scores of Black gymnasts.

Running alternate specifications of our model with different percentile cutoffs for outliers returned a variety of results. In the most interesting of those specifications, there were statistically significant coefficients on our interaction term in full fixed-effect models of the uneven bars. Those specifications showed that Black gymnasts scored lower on the uneven bars when competing at BYU to an extreme and alarming degree; one model even suggested that Black gymnasts lost *an entire point* on average when they performed at BYU. We later discovered an error in the data cleaning process whose correction eliminated these results. We share this anecdote to highlight how sensitive our model is to the small size of our sample of

scores received by Black gymnasts while performing at BYU; after all, only 1.3% of our sample is composed of scores received by Black gymnasts at BYU (see Table 1). If we were to see the same distributions of scores reflected in Figure 1 in a better-balanced sample of similar size, our model would likely reveal practically large and statistically significant effects. As it stands, this is not the case, and our analysis is inconclusive.

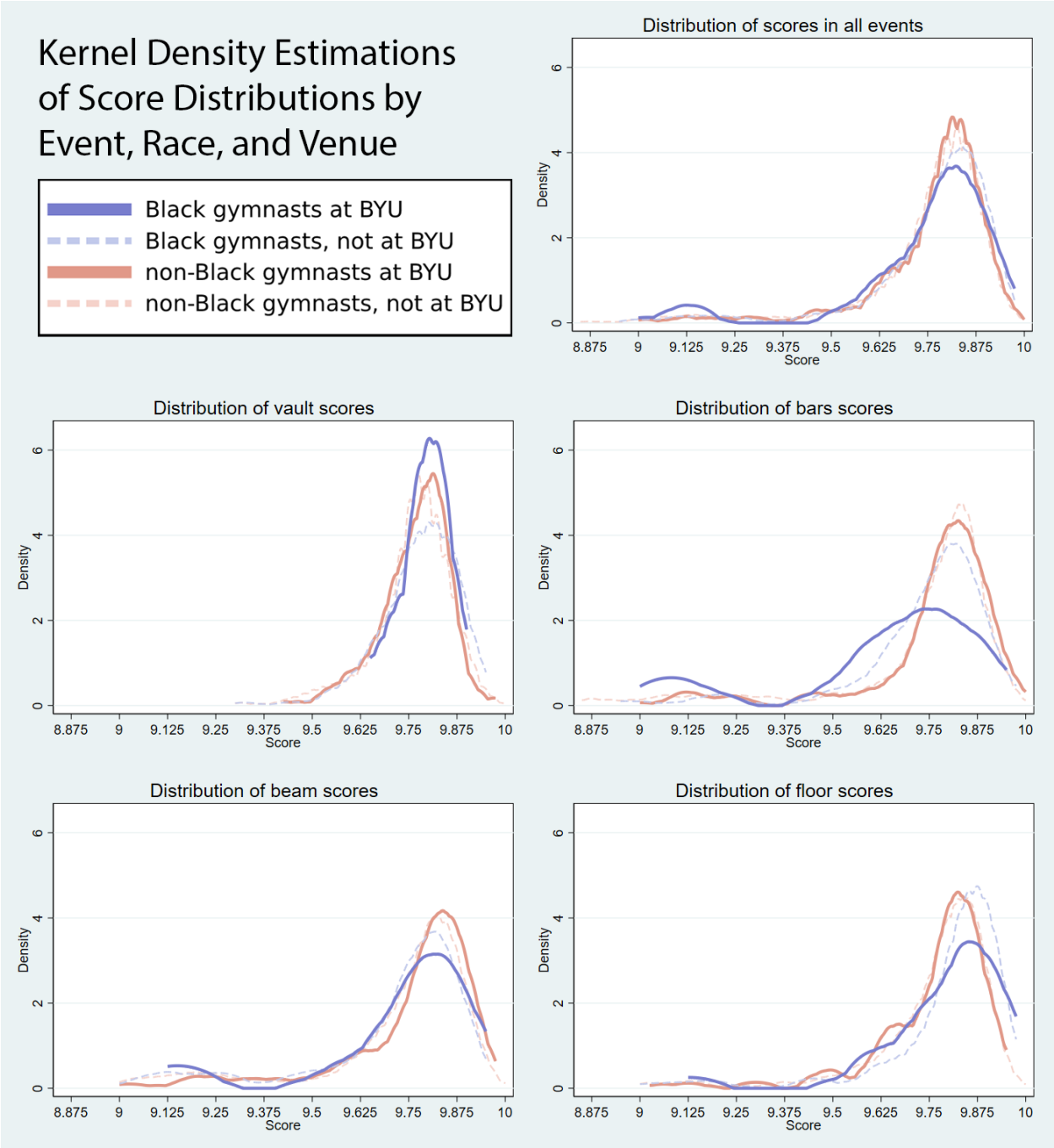
Section V. Conclusion

Using a unique dataset on collegiate women's gymnastics scores composed of teams that visit BYU, we test whether Black gymnasts experience a change in performance when competing at BYU that their non-Black competitors do not experience. Our data allows us to introduce many sets of fixed effects to control for relevant factors that influence scores, which allows us to focus on the interaction between a gymnast's race and the venue at which they perform. We find that there is no significant difference in score distributions at BYU between Black and non-Black gymnasts in any individual event or overall within our sample.

Would our findings have been different with a larger or more balanced sample? We note that the relative scarcity of Black gymnasts' BYU scores in our dataset limits our ability to parse out any heterogeneous effect that may exist between Black and non-Black gymnasts at BYU. This limitation restrains us from concluding that Black gymnasts are harmed by intergroup anxiety or other similar social pressure, but it also restrains us from concluding the opposite. If teams with more Black gymnasts begin to visit BYU more frequently in the future, or if we study meets at schools that have recently had prominent race-related scandals in their gymnastics programs, such as Alabama or UCLA, we may later reach a stronger conclusion.

FIGURE 1

KERNEL DENSITY ESTIMATIONS OF SCORE DISTRIBUTIONS



Notes. This figure only reports scores that are above the 3rd percentile of scores in each event in an effort to diminish the effect of outliers. This means scores in vault are 9.3 or higher; scores in bars are 8.85 or higher; and scores in both beam and floor are 9 or higher. We felt this percentile measure struck a good balance between maintaining reasonable score cutoffs across events and removing outliers.

TABLE 1
SUMMARY STATISTICS

VARIABLES		(1) Vault	(2) Bars	(3) Beam	(4) Floor	(5) All Events
Score	Mean	9.765	9.726	9.719	9.766	9.748
	(SD)	(0.099)	(0.223)	(0.213)	(0.164)	(0.179)
% Black		0.171	0.133	0.114	0.135	0.138
		(0.376)	(0.340)	(0.318)	(0.342)	(0.344)
% at BYU		0.098	0.100	0.101	0.099	0.099
		(0.297)	(0.300)	(0.302)	(0.298)	(0.299)
% Black and atBYU		0.015	0.013	0.010	0.013	0.013
		(0.122)	(0.115)	(0.100)	(0.115)	(0.114)
Observations		1859	1873	1879	1863	9337

Notes. All statistics reported in Columns 1-4 were calculated after excluding the bottom third percentile of scores in that event. Statistics in Column 5 were calculated from the cumulative dataset represented by Columns 1-4. The % variables represent the fraction of scores in a given event that were received by a gymnast in that variable category.

TABLE 2
EFFECTS OF GYMNAST RACE ON PERFORMANCE AT BYU

		Coefficient on the (Black x atBYU) interaction term				
MODELS		(1) Vault	(2) Bars	(3) Beam	(4) Floor	(5) All Events
Simple OLS on Black, atBYU, and interaction	Coefficient	0.009	-0.141***	-0.013	-0.013	-0.035*
	(Std. Error)	(0.013)	(0.047)	(0.053)	(0.039)	(0.020)
Introduce Gymnast and Venue effects		-0.001	-0.087**	-0.022	-0.024	-0.031
		(0.014)	(0.043)	(0.063)	(0.040)	(0.020)
Add Year and Judge effects		0.017	-0.062	-0.011	-0.009	-0.027 †
		(0.013)	(0.043)	(0.071)	(0.046)	(0.021)
Add Gymnast-by-Year and Judge-by-Race effects		0.022	-0.088	-0.036	-0.032	-0.035 †
		(0.105)	(0.276)	(0.147)	(0.076)	(0.033)
Observations		1,859	1,873	1,879	1,863	9,337
R-squared, by Model		0.005	0.005	0.003	0.005	0.002
		0.528	0.309	0.314	0.424	0.225
		0.631	0.404	0.432	0.539	0.311
		0.709	0.479	0.519	0.613	0.373

*** Significant at the 1% confidence level
** Significant at the 5% confidence level
* Significant at the 10% confidence level
† Also includes Event fixed effects

References

Britt, T. W., Bonieci, K. A., Vescio, T. K., Biernat, M., & Brown, L. M. (1996). Intergroup Anxiety: A Person × Situation Approach. *Personality and Social Psychology Bulletin*, 22(11), 1177–1188. https://doi.org/10.1177/01461672962211008

Garicano, Luis and Palacios-Huerta, Ignacio and Prendergast, Canice (2001). Favoritism Under Social Pressure. *NBER*, Working Paper No. w8376, Available at SSRN: https://ssrn.com/abstract=276006

National Collegiate Athletic Association [NCAA] (2018). NCAA Demographics Database. *NCAA.org*. https://www.ncaa.org/sports/2018/12/13/ncaa-demographics-database.aspx

Price, Joseph and Wolfers, Justin (2010). Racial discrimination among NBA referees. *The Quarterly Journal of Economics*, 125(4), 1859-1887. https://users.nber.org/~jwolfers/papers/NBARace(QJE).pdf

Stephan, Walter G., Stephan, Cookie White. (1985). Intergroup Anxiety. *Journal of Social Issues*, 41(3), 157–175. https://doi.org/10.1111/j.1540-4560.1985.tb01134.x